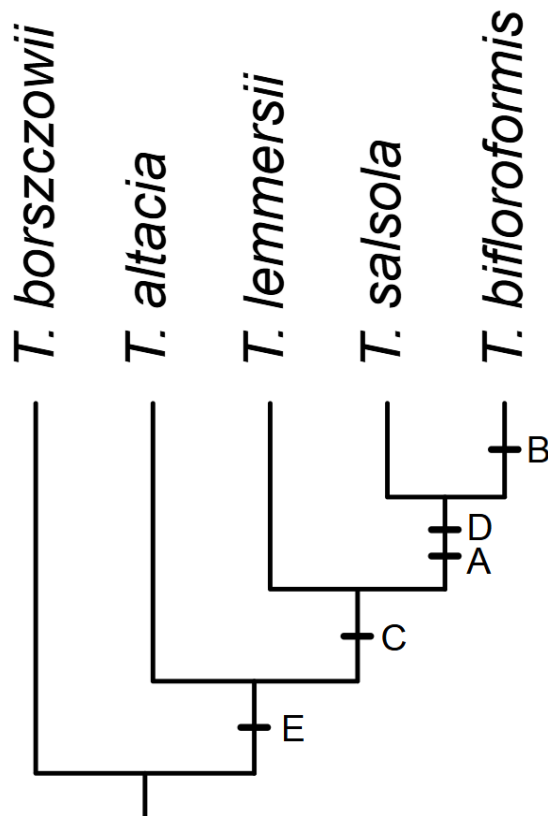


Theory Part A

Biosystematics

Task #1.

- A) **FALSE.** checked Explained by a figure.
- B) **FALSE.** checked Explained by a figure.
- C) **TRUE.** checked They are monocotyledons.
- D) **TRUE.** checked Explained by a figure.



Task #2.

- A) **TRUE.** checked By employing deductive reasoning and systematically listing options, one can logically identify each of the five owl species displayed on the table. If the process is executed accurately, it becomes evident that the fifth species belongs to the *Surnia* genus (*Surnia ulula*), which is the sole representative of its kind.
- B) **FALSE.** checked The second most related pair is 3/5 with 2. (1 - *Asio*, 2 - *Athene*, 3 - *Glaucidium*, 4 - *Otus*, 5 - *Surnia Ulula*).
- C) **FALSE.** checked All species belong to different genera.
- D) **TRUE.** checked If the most recent common ancestor of the *Pulsatrix-Meagoscop* clade lived in Africa / Eurasia, that would require 2 events of migration to America as well as two events of extinction in Africa and Eurasia. In the

opposite case (common ancestor lived in America), we need to assume only a single migration from America to Africa / Eurasia.

Task #3.

- A) **TRUE.** checked. The most parsimonious solution for alignment I is the topology depicted on phylogenetic tree 1 (5 changes across the tree in total). Note, it is given that D is an outgroup.
- B) **FALSE.** checked. Sites 3 and 5 are not useful. Site #3 is identical for all taxa. While site #5 has one variation - taxon C has A (not T), it is not helpful for the phylogenetic reconstruction via parsimony - you just need to make one change that leads to the taxon C ($T \rightarrow A$) and it'll be the most parsimonious solution for any tree topology.
- C) **FALSE.** checked. Phylogenetic tree 1 is incongruent with the reference tree, while RNA polymerase β is a known conserved gene in Bacteria compared to transposase gene that is part of the transposons.
- D) **FALSE.** checked. As species D is an outgroup, the remaining three taxa can be phylogenetic related in three possible ways - D,A,(B,C); D,B,(A,C); D,C,(A,B)

Cell and Molecular Biology

Task #4.

- A) **FALSE**(Checked). Since oxygen consumption is higher in tumor cells, we can conclude that they have elevated metabolism. They need to consume more oxygen because of a relatively high proliferation rate.
- B) **TRUE**(Checked). The differences in extracellular pH values can be explained by abnormally elevated rates of anaerobic glycolysis in tumor cells, which results in higher production and accumulation of lactic acid. This phenomenon is known as the Warburg effect.
- C) **FALSE**(Checked). Figure 2 demonstrates that the oxygen consumption rate is elevated in tumor cells. Such a high demand can't be satisfied, which leads to the development of hypoxia in the tumor microenvironment.
- D) **FALSE**(Checked). Error bars of the data in Figure 2 demonstrate that the cancer cells of patient 2 display the most variation in the single-cell OCR. Hence, they are more heterogeneous compared to the cells of other patients.

Task #5.

- A) **TRUE**. Checked Tyr-33 substituted for aromatic amino acids such as tryptophan and phenylalanine shows higher affinity values than for other mutants.
- B) **FALSE**. Checked In Figure 1A.1, it can be seen that the original amino acid Asn has a higher affinity to all cardiac glycoside molecules than all mutants.
- C) **TRUE**. Checked The substitutions Tyr-33-Trp and Trp-100-Arg show the highest absorbance ratios in their mutations. Therefore, they can be used for cardiac glycoside detection systems.
- D) **TRUE**. Checked Digoxigenin lacking sugar in R1 behaves differently from other glycosides.

Task #6.

- A) **FALSE**. Checked The structure (a) in Figure 1D has Φ and ψ angles of about -100 and 120 degrees, respectively. It is mostly located in Region 1 in Figure 2.
- B) **FALSE**. Checked The structure (b) in Figure 1D has Φ and ψ angles of about -60 and -60 degrees, respectively. It is mostly located in Region 2 in Figure 2.
- C) **TRUE**. Checked The structure (c) in Figure 1D has Φ and ψ angles of 90 and -90 degrees, respectively. From the Ramachandran plot, it can be seen that amino acids with Φ and ψ angles of 90 and -90 degrees are uncommon.
- D) **TRUE**. Checked From the Ramachandran plot, it can be seen that amino acids with Φ and ψ angles of 0 degrees are not present in the rabbit's pyruvate kinase. They are prohibited by steric interference between atoms in the polypeptide backbone and amino acid side chains.

Task #7.

- A) **TRUE.** Checked Actively dividing cells that need R5P for nucleotide synthesis. The reactions catalyzed after the first TKT-catalyzed reaction use R5P, which is essentially needed for cells to divide fast. Therefore, only the reactions before the first TKT-catalyzed reaction are more active (than the reactions occurring after the first TKT-catalyzed reaction) in fast-growing cells.
- B) **FALSE.** Checked From the reaction pathways, it can be seen that TKT transfers two-carbon fragments.
- C) **TRUE.** Checked With G6PDH deficiency less NADPH will be produced. Also, ROS will be reduced by existing NADPH, leading to a decreased level of NADPH.
- D) **FALSE.** Checked From the reaction scheme, it is seen that for the production of two molecules of F6P three G6P are needed.

Task #8.

- A) **FALSE.** (Checked) Pictures A and B show that all the bases of all loops are in the same place - therefore each of the CTCFs participates in more than one loop. Picture A shows formation of 2 or 3 loops, if each CTCF-binding site would only participate in one loop, maximum number of loops would be 2.
- B) **FALSE.** (Checked) DNA methylation prevents protein binding (CTCF in this case), thus only 2 loops are formed instead of 3. The site by which a new loop is formed corresponds to the 3rd CTCF site.
- C) **TRUE.** (Checked) Expression activity of IGF2 depends on the proximity of enhancer, figures A and B imply that this enhancer is located between H19 gene and CTCF binding site 4. Expression of H19 depends on the methylation state of ICR - it is located close to its promoter and in the methylated state transcription factors cannot access DNA.
- D) **TRUE.** (Checked) As was mentioned in the previous explanation, expression of H19 depends on the methylation state of its promoter, which overlaps with the ICR, regardless of the enhancer.

Task #9.

- A) **TRUE.** (Checked) Since cryo-EM works by averaging many low-resolution images of molecules, a region that is different among these images will be not clear in the final density map and that means that this region is more flexible or has more stable conformations.
- B) **FALSE.** (Checked) From the molecular model, we can see that these helices are holding subunits of ATAD1 together. One side of the helix should have hydrophobic residues to interact with another subunit, another side should be hydrophilic to interact with water.
- C) **FALSE.** (Checked) The specificity of the substrate is mainly affected if the central pore has mutations, we see that E193Q is clearly outside of the pore.

- D) **FALSE.** (Checked) ATAD1 indeed translocates proteins, it can be deduced from the motion of M6 and overall structure, however, it pulls mislocalized transmembrane proteins out of the membrane. This conclusion is made from fluorescent assay (Gos28 is in the membrane) and structure of pore (aromatic amino acids).

Task #10.

- A) **FALSE.** (Checked) The shortest possible path is movements along a vertical line, the real benefit of having magnetosomes is in energy efficiency - bacteria are not spending any energy to align themselves on path towards OATZ (even if this path is not the most efficient).
- B) **FALSE.** (Checked) Since the polarity of cells is opposite on both hemispheres, the movement of flagella will be the same for bacteria from the southern hemisphere.
- C) **TRUE.** (Checked) Since axial magnetotactic bacteria do not follow the direction of the field they have two options for movement - to the left and the right.
- D) **FALSE.** (Checked) The situation describing the beginning of the experiment is bacteria below a certain depth level in figure B. As we can see bacteria move and rotate their flagella clockwise. In the setup C there is no oxygen in the middle and concentration increases towards the ends.

Task #11.

- A) **FALSE.** (Checked) Library construction is necessary to make variance, which is one of the important criteria for natural/artificial selection.
- B) **FALSE.** (Checked) Anaerobic conditions for cultivation are necessary to avoid usage of electron transport chains for the NAD⁺ regeneration.
- C) **TRUE.** (Checked) The only connection of this heterologous reaction catalyzed by Enzyme D to native metabolic network is NAD⁺/NADH usage. Otherwise, the selection pressure cannot affect the enzyme D.
- D) **TRUE.** (Checked) As the oxygen concentration in the large intestine is very low, the strains having higher Enzyme D activity have higher NAD⁺ regeneration rate at the anaerobic conditions, which is the case in the large intestine.

Task #12.

- A) **TRUE.** (Checked) DMSO₂ is a less hydrophilic molecule than DMSO, and it is stable compared to DMSO₄ (which rapidly hydrolyses in water).
- B) **TRUE.** (Checked) DMSO is a strongly hydrophobic compound used as a solvent for organic molecules and an agent facilitating penetration of organic membranes, including skin.

- C) **FALSE.** (Checked) Gibbs free energy (which includes enthalpy) influences the feasibility of the reaction, but the reaction rate is the function of the activation energy.
- D) **FALSE.** (Checked) Methastases block most of the elimination of the DMSO and its derivatives, however, as was mentioned in the text odor was observed indicating elimination through the lungs.

Ecology

Task #13.

- A) **FALSE.** Checked Both NK+ graphs for legume cover are marked with a dot, which means they are both statistically significant.
- B) **FALSE.** Checked Biomass graphs only differ in magnitude between 3 years and 6 years.
- C) **TRUE.** Checked Every time there is added nitrogen, the legumes cover, species richness, or biomass on average decline. That is not true for other nutrients, in which legumes do not have a competitive advantage of fixation. Therefore, steppes must indeed be nitrogen-limited, as taking away nitrogen advantage lead to negative effects for the species with that advantage.
- D) **TRUE.** Checked Non-nitrogen fixers do not benefit from low soil nitrogen, so any additional nutrients will drive their growth.

Task #14.

- A) **TRUE.** (Checked) According to the hypothesis C, the tropics have a higher range of ages which would increase the variance in the species age. The temperate zone has less variance due to their later starting point of diversification.
- B) **TRUE.** (Checked) This means that a pattern has been established regardless of the starting point of diversification caused by the global catastrophe.
- C) **FALSE.** (Checked) Greater seasonal variation is in the temperate zone. And if it increases the diversification rate in the temperate zone this would go against Hypothesis B.
- D) **TRUE.** (Checked) This implies that all taxa emerged initially in the tropical regions and specific diversification of temperate species started after the start of the tropical diversification.

Task #15.

- A) **TRUE.** (Checked) When warm and moist air becomes cold, excess moisture is released as precipitation, and increased precipitation leads to larger gross primary production.
- B) **FALSE.** (Checked) It is La Niña because it will cause an increase in precipitation over Australia (western Pacific region).
- C) **FALSE.** (Checked) During El Niño, wind over the Pacific decreases, and the depth of transition between warm and cold water increases, due to this upwelling decrease.
- D) **FALSE.** (Checked) Changes in year Y are caused by a decrease in salinity of a lake which is caused by increased precipitation. Increased precipitation over North America is possible when El Niño happens.

Task #16.

- A) **FALSE. (Checked)** Here slope depends on d in the equation, since d is the complexity of the pattern it will have a larger value for rough coastlines
- B) **FALSE. (Checked)** Since the graph is on a logarithmic scale, there will be more individuals with smaller scales than individuals of average species, even if there are only half of them.
- C) **FALSE. (Checked)** This scale depends on the resources or territory required for each individual, not the size of the individual.
- D) **TRUE. (Checked)** In herds, the perspective scale of each individual would depend on the overall perspective scale of the entire herd.

Task #17.

- A) **TRUE. (Checked)** Water withdrawal increases the speed of water circulation and water molecules travel faster among water reservoirs
- B) **FALSE. (Checked)** While high Rws always represent a water crisis, low Rws can be explained by a water crisis caused by an inability to utilize available RFWR (large Q and small W).
- C) **TRUE. (Checked)** Map with $1/Aw$ will have weighted values of population and water availability and thus some regions with small population density would have opposite values in $1/Aw$ and Rws
- D) **TRUE. (Checked)** Food and industrial goods production requires much more water than is required for consumption by humans and thus represents a major water

Ethology

Task #18.

- A) **TRUE.** (Checked) In the case of Western Barn Owls, failing to respond to potentially lethal warnings is perhaps because they are used to hunting, not getting hunted, and alarming sounds such as approaching vehicles are new to them.
- B) **FALSE.** (Checked) The speeding behavior allows them to exploit traffic gaps for a period when traffic volume is low. The plateau represents a steady increase of probability of mortality associated with the speeding behavior.
- C) **TRUE.** (Checked) When traffic has reached sufficient volume for an animal to pause before attempting crossing, the probability of avoidance becomes greater than the probability of mortality.
- D) **FALSE.** (Checked) Even slight traffic volume can restrict movement of avoiders leading to the lowest mortality among others.

Task #19.

- A) **TRUE.** (Checked) Information about effective routes is passed from generation to generation
- B) **TRUE.** (Checked) In each generation of the experimental group, each new pigeon requires time to learn the route before showing actual performance
- C) **TRUE.** (Checked) If it would not, there will be no increase in efficiency over generations in the experimental group, which can be explained by “innovations” introduced by a new pigeon in a pair
- D) **TRUE.** (Checked) Collective decision making increases with the number of animals in the group, but figures show clear difference between experimental and control pair performance, while both of them contain 2 pigeons

Genetics & Evolution

Task #20.

A) **FALSE. Checked**

p: A1

q: A2

r: A3

$$p+q+r = 1$$

$$p^2+q^2+r^2+2pq+2pr+2qr = 1$$

$$A3A3 = r^2 = 0,01; \quad r = 0,1$$

$$A2_ = q^2+2qr = 0,08$$

$$q^2+2qr+0,08 = 0; \quad q = 0,2$$

$$\begin{aligned} ax^2 + bx + c &= 0 \\ x_{1,2} &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$$

$$r = 0,1; \quad q = 0,2; \quad p = 0,7$$

So individual in n generation make up 70%

B) **TRUE. Checked** A2A2 and A2A3

C) **FALSE. Checked** Meiosis does not ensure the distribution of two sets of chromosomes (paternal and maternal) at 50% in the haploid daughter cell.

D) **FALSE. Checked** A lethal mutation most likely will not be able to pass through the haploid generation, since it will not be covered by a dominant allele

Task #21

A) **FALSE. checked.** More genetically diverse populations have larger effective population sizes.

B) **TRUE. checked.** As the African population has higher genetic diversity, the deleterious (or slightly deleterious) mutations are less likely to get fixed (frequency = 1) as the selection plays greater role comparatively to the genetic drift.

C) **TRUE. checked.** Since humans started expansion out of Africa, different geographic regions were colonized by different population groups. As a result, all new populations were established by a small number of people from a larger (African) population.

D) **FALSE. checked.** As human genomes are almost identical to each other, including conserved DNA polymerases, the opposite is true and the heterozygosity is best studied in highly variable regions, such as microsatellites.

Task #22.

- A) FALSE(Checked). The high consumed amount of exogenous ethanol is a consequence of agricultural output i.e. cultivation of certain crops and their intended fermentation, therefore, prior to the extensive development of agriculture, ADH was adjusted to degrade only tiny amounts of endogenous or exogenous alcohol.
- B) TRUE(Checked). The adverse effect of acetaldehyde accumulation in the body is going to discourage individuals from ethanol consumption.
- C) FALSE(Checked). There is a lack of information to claim that, but the evident low frequency of ADH1B*2 gives a clue for absent ethanol consumption-driven positive selection.
- D) TRUE(Checked). With the given information, the assumption of positive selection geographically correlates with the presence of supposed selective force.

Task #23.

- A) **FALSE**. checked. The frequency of A1 converges to the ~0.3-0.35 and remains relatively stable, suggesting that selection favors the intermediate genotype (stabilizing selection). The tricky part of the question is that while frequency of A2A2 > A1A2, the relative fitness of A2A2 < A1A2 that is why the populations converge to stable equilibrium. If relative fitness of A2A2 > A1A2, then allele A1 would go extinct over time.
- B) **FALSE**. checked. As the random fluctuations from generation to generation are more pronounced, it suggests that Aizhan decreased the population size in experiment #2 (genetic drift starts playing a more important role, increasing overall stochasticity due to the random chance).
- C) **TRUE**. checked. Almost all populations (including the theoretical fit) reach some degree of the stable equilibrium, suggesting the heterozygote advantage (it can be actually easily derived from Hardy-Weinberg equilibrium).
- D) **FALSE**. checked. The frequency of A1 decreases below 0.5 before reaching some degree of stability, indicating that relative fitness of genotypes is the following one: A1A2>A2A2>A1A1

Task #24.

- A) **TRUE**. checked. Ribosomal proteins are essential and universally highly expressed, as translation is the fundamental cellular process. Hence, ribosomal genes tend to have very high codon usage bias to synthesize ribosomal proteins rapidly and efficiently by utilizing the most abundant tRNA pool.
- B) **TRUE**. checked. As codon usage bias tends to improve the translational efficiency, correction of the transgenic DNA to account for that phenomenon generally leads to more efficient expression of it by the recipient's organism.

- C) **FALSE.** checked. Viruses tend to have a pronounced codon usage bias. Furthermore, even when viruses don't carry their tRNA molecules, they still rely on their host translational machinery and tRNA. Hence, genetic material of viruses will be selected to fit their host tRNA pool. In addition, multiple phages (bacterial viruses) can also carry their own tRNA molecules to account for their codons content. (multiple explanations possible, not only tRNAs)
- D) **FALSE.** checked. Methionine cannot have codon usage bias as it is made only from one codon.

Task #25.

- A) **FALSE.** checked As it can be seen from the figure 2b, the mutation X is located near to the amino acids that are less conserved rather than the mutation Z, which is located in a highly conservative region. As the active site of an enzyme is crucial for its function, the active sites are likely to be more conserved.
- B) **FALSE.** checked The genetic code for Threonine (Thr) is AC_, and for Alanine (Ala), it is GC_. Assuming single-nucleotide changes in the mutations, the deduced mutation of A → G leads to the genetic code GC_ for Thr. Given that both A and G are purines, this mutation is identified as a transition (purine → purine or pyrimidine → pyrimidine), not a transversion (A/G → T/C).
- C) **FALSE.** checked It can be observed that the two subunits possess MTS, indicating that they are transported into the mitochondrion separately. This suggests that the assembly of RNase takes place within the mitochondrion rather than externally before transport.
- D) **TRUE.** checked We know that individual II-3 (A1) is the carrier of mutation X. We need to find the probability that individual II-2 (A2) is a carrier of mutation Y, which is 2/3. The probability of them having a child with the syndrome is 1/4. Therefore, the overall probability of them having a child with the syndrome is $2/3 * 1/4 = 1/6$.

Task #26.

- A) **TRUE.** (Checked) Since it is not crossing the threshold line and we see a much longer duration, we can conclude that the nature of the event at the Frasnian-Famennian boundary (FF on the graph) is not the same as all other mass extinctions.
- B) **FALSE.** (Checked) Based on smaller events graph can be biased, however absence of large recent events supports the general trend.
- C) **TRUE.** (Checked) Events above that line mean large mass of carbon added in a short period of time, almost all events over that line are mass extinctions
- D) **TRUE.** (Checked) Since there are more species and they are more efficient, carbon travels faster between reservoirs, and carbon perturbation events happen in a shorter time scale.

Task #27.

- A) **TRUE. (Checked)** The efficiency of E104K and G238S increases significantly when they are combined.
- B) **FALSE. (Checked)** Cumulative probability plot is skewed to the right, meaning that trajectories have decreasing probabilities with trajectory 1 being the most probable.
- C) **FALSE. (Checked)** Phenotypic fitness does not influence mutation rate but increases the probability of mutation fixation.
- D) **FALSE. (Checked)** If there will be intermediate form with high fitness and next forms have decreasing fitness, then this route is not always the most probable, regardless of the final fitness. If there will be local fitness maxima evolution would not favor further mutations decreasing fitness (evolution cannot plan).

Task #28.

- A) **TRUE. (Checked)** Effectively neutral mutations, while may not be truly neutral, have negligible effect on the scales of real populations (usually $> N_e$) $\Rightarrow$ their fixation can be explained only by random genetic drift. For example, suppose, $s < 1/N_e$, e.g. humans ($N_e = 10000$) $\Rightarrow s < 1 \cdot 10^{-4}$ (meaning that mutation has a very weak effect so that it is effectively neutral (cannot be efficiently selected within the actual population - its fate completely depends on drift)
- B) **TRUE. (Checked)** Species with larger N_e have less effectively neutral mutations and consequently larger amounts of weakly adaptive mutations.
- C) **TRUE. (Checked)** We are not able to track one large group of deleterious mutations - lethal mutations since only survivors are passing their mutations.
- D) **FALSE. (Checked)** It represents the proportion of effectively neutral mutations, when we compare genomes deleterious mutations should be under negative selection and not present in large amounts - we mostly see effectively neutral mutations at non-silent sites.

Task #29.

- A) **TRUE. (Checked)** Only infection of germline cells will generate heritable genetic element
- B) **FALSE. (Checked)** Loss of one gene, for example - env, is enough to lose the property of virion formation and to become a retrotransposon
- C) **FALSE. (Checked)** Since LTR regions are similar, they are prone to homologous recombination events, where everything between them is lost
- D) **TRUE. (Checked)** This factor would prevent certain retrovirus from infecting the host and when the exogenous retroviral population becomes too small or extinct, then the population no longer needs this ERV in their genome

Plant Anatomy and Physiology

Task #30.

- A) **FALSE.** **checked** From Figure 1, it's evident that when a shoot from a wild-type plant is grafted onto the root of a gibberellin-deficient plant, there are no observable changes. However, when a *ga1-3* shoot is grafted onto a *ga1-3* root, there's a decrease in intensity, indicating that gibberellin synthesis occurs in the shoots and is transported to the roots.
- B) **TRUE.** **checked** Examining Figure 2 reveals that the wild type exhibits a high concentration of gibberellin in the vacuoles of the pericycle and a relatively low concentration in the endoderm. This pattern is consistent across all mutants except the last one, where the ability to store gibberellin in the vacuole is compromised. Notably, in this mutant, the cytoplasmic concentration of gibberellin in the endoderm and cortex is significantly higher than that of the wild type. This indicates that gibberellin, instead of being stored in the vacuole, is being transported into the endodermal cells, resulting in an elevated concentration. This observation underscores the regulatory role of vacuoles in gibberellin secretion into the endoderm and cortex.
- C) **TRUE.** **checked** This mutant exhibits impaired gibberellin transport into the endodermis, resulting in a low level of suberization in these cells. Since endodermal suberization is crucial for the formation of Casparian strips, this mutant will have poorly defined Casparian strips. This will lead to an increase in apoplastic transport into the stele, which increases the chances of obtaining water imbalance or toxin contamination.
- D) **FALSE.** **checked** The transporter *npf 2.14* is located on the vacuolar membrane. This can be observed in Figure 2, where its absence resulted in a reduction of vacuolar gibberellin concentration to zero.

Task #31.

- A) **TRUE.** Checked Because GDC, a decarboxylase, removes a CO₂, but does not prevent serine production. Serine can go into gluconeogenesis.
- B) **FALSE.** Checked Increased CO₂ levels will lead to Rubisco favoring photosynthesis over photorespiration, so LESS need for the latter.
- C) **FALSE.** Checked All enzymes there are either pink or yellow, so overexpressing them would have a negative net effect.
- D) **TRUE.** Checked Because their Calvin cycle happens in cells with high concentration of CO₂.

Task #32.

- A) **TRUE.** Checked Because electrodes could trigger hyperpolarization or depolarization, bypassing light stimuli and causing stomatal opening/closing.

- B) **FALSE.** Checked Lignin makes cells more rigid, preventing subsidiary cells from properly "bending" to allow a wide opening, thus making stomata narrower and less effective.
- C) **FALSE.** Checked Potassium deficiency would lead to lower volume of water flow, as potassium gradients would have lower magnitude, so the cell shape would change less and pore area would be lower
- D) **TRUE.** Checked Since aquaporins are passive channels, they cannot start or end a flow on their own, so adding more of them could only increase the rate of osmosis.

Task #33.

- A) **TRUE.** Checked Without the xanthophyll cycle, the plants cannot dissipate excess light's energy. Therefore, they will suffer strong oxidative stress under the sun, and would prefer shade.
- B) **TRUE.** Checked The Steppe population is constantly exposed to direct sunlight, while the understory population is hidden by the forest canopy. Therefore, the former would be more adapted to constant daylight, partially through higher xanthophyll concentration. As it is a temperate climate, extra heat does not make this a maladaptive trait.
- C) **FALSE.** Checked High leaf temperature means that using heat as an energy dispersion mechanism is more dangerous, and just reducing de-epoxidase activity would not help as under sunlight the proton concentration would increase regardless, activating more zeaxanthin. Thus, total Pre-Z + Z concentration should be decreased for the plant to survive. Since the climate is always sunny and hot, it is a viable adaptation.
- D) **TRUE.** Checked This is a hypothetical scenario, but if no other option was available, the leaves from the top of the canopy would have to produce more ascorbate (salt of ascorbic acid, also known as vitamin C) to adapt to higher light levels. Thus, eating them gives more vitamin C. From the book: "The pool of reduced ascorbate, which plays a role in the xanthophyll cycle (Fig. 2.17), is also several-fold greater in plants acclimated to high light (Logan et al. 1996)." page 45

Task #34.

- A) **FALSE.** Checked By 9am, both starch and soluble sugar graphs had positive change for all three stages. Therefore, photosynthesis must have begun before that, as it is the sole source of production of new starch and soluble sugars for photosynthetic cells.
- B) **TRUE.** Checked Peak value of 120 instead of 72-90 is unlikely to come due to grains growing the seed. TSC is easier to transport through phloem towards the seeds compared to starch, so there is an increased demand for TSC in seed kernels.

- C) **TRUE.** Checked Since the TSC pool includes free cytosolic glucose, which is the fuel for cell respiration, this should be true. Between 24 and 6 hours, that glucose is going to be respired further, reconnecting the graphs.
- D) **FALSE.** Checked Lower leaf starch means more carbohydrates are leaving leaves to go to the seed's endosperm. Since it is a crop, humans will eventually consume endosperm and thus, more starch going to the endosperm is a beneficial trait.

Task #35.

- A) **TRUE.** Checked Density is inversely proportional to PC3, and *S. giganteum* has lower PC3 values than the other species.
- B) **FALSE.** Checked Resistance to xylem blockage is tracked by PC2, and height is tracked by dot size. For *S. sempervirens*, dot size and PC2 are positively correlated, so it is the taller trees that would have higher resistance to xylem blockage
- C) **TRUE.** Checked In an arid environment—adapted plants, controlling transpiration through changes in leaf shape and structure is a more advantageous tactic to costly branch-level changes to xylem structure. Therefore, the change of the environment to a wetter one will allow the plants to pursue the costlier strategy.
- D) **FALSE.** Checked To compare density and height, we compare the dot size and PC3. Bigger dots tend to be close to 0.5 mark for PC3, and smaller dots cluster at higher PC3. However, they also intermix with big dots near 0.5, so PC3 (density) and dot size (height) don't appear particularly covarying at all. Furthermore, density inversely covarying with height does not make biological sense, as taller trees would want denser, more flexible and taut stems.

Task #36.

- A) **FALSE.** Checked Since diffusion coefficient in water is smaller than air, the aquatic plants prefer thinner cuticles to allow more area to uptake/release the dissolved gasses, as opposed to only stomata of land plants.
- B) **TRUE.** Checked The graph between 7 and 8 shows that HCO_3^- is at least 90% of total carbon in that range of pH, so it would be advantageous to use it as a carbon source in oceanic ecosystems.
- C) **TRUE.** Checked True, because it would locally lower the pH to levels where CO_2 dominates and can be more readily uptaken as opposed to its ionic forms.
- D) **TRUE.** Checked Higher efficiency Rubisco would lead to higher forward reaction rate, which would create the asymmetry to give direction to carbonic anhydrase reaction. This assumes the pH stays close enough to its original value, which is possible because cytoplasm is buffered and anhydrase does not consume enough protons to shift the pH too far.

Task #37.

- A) **FALSE. Checked** This refers to the Casparian belts, which do not allow water and solutes to pass through the cell walls (apoplastic pathway), only through cell membranes. Casparian belts are located in the endoderm, that is, on the border of zones B and C.
- B) **FALSE. Checked** In order for water to flow from the soil (zone A) to the primary root cortex (zone B), a solution with a higher concentration of dissolved substances must be formed in zone B (that is, in zone B the water potential must be lower). Solutes combine with water molecules due to their charge, reducing the concentration of free water molecules. Thus, for the shown water flow, it is necessary that the concentration of free water molecules in zone B be less than in zone A.
- C) **TRUE. Checked** Oxygen starvation usually occurs during floods. As a result of the described process, empty cavities appear in zone B (primary root cortex), which improves the aeration of root tissues.
- D) **TRUE. Checked** True, because NaCl dissociates in water into two particles. That is, 0.2 moles of NaCl yield 0.4 Osm/L in solution. Thus, the solution will have a higher concentration of dissolved particles than the cells in zone B (i.e., it will be more concentrated).

Animal Anatomy and Physiology

Task #38.

- A) **FALSE.** Checked The SA node should have its own pace - C3. C1 corresponds to the ventricular myocyte and C2 to the atrial one.
- B) **FALSE.** Checked Repolarization of atria is not visible behind the QRS complex, it would be seen if there is no depolarization of ventricles.
- C) **FALSE.** Checked S wave is depolarization, but moving in the opposite to the lead side.
- D) **TRUE.** Checked The current direction of the vector P corresponds to the direction of the lead II and is perpendicular to the lead III.

Task #39.

- A) **FALSE.** Checked Snow leopards' hemoglobin shows almost no sensitivity to DPG, as evidenced by nearly identical curves for cofactor-free hemoglobin and hemoglobin with added DPG, unlike the noticeable difference seen in humans between cofactor-free and DPG-added hemoglobin (Figure 1B).
- B) **TRUE.** Checked The hemoglobin of snow leopards inhabiting high-altitude areas has a low affinity to oxygen and causes hypoxia. An enhanced capacity to increase ventilation under severe hypoxia could help compensate for the low O₂ affinity of snow leopards, causing respiratory alkalosis. This, in turn, induces increased renal excretion of bicarbonate compared to snow leopards living in lowland areas, returning the pH back to normal.
- C) **FALSE.** Checked From Figures 1A and 1B, it can be observed that the oxygen affinity of hemoglobin in snow leopards is mostly the same as in lions for low oxygen concentrations, which inhabit lowland areas. Therefore, the hemoglobin of snow leopards is not adapted to high altitudes biochemically, and snow leopards rely on other adaptations like hyperventilation.
- D) **FALSE.** Checked Increased P₅₀ means that the hemoglobin has less affinity to oxygen. The graph shows a positive change in P₅₀, which implies that KCl always lowers affinity. Therefore, KCl causes a stronger decrease in affinity to oxygen for the lion's HbA as opposed to the snow leopard's HbB (Figure 1C).

Task #40.

- A) **TRUE.** Checked From Figure 1, it can be observed that the light-fed mice were consistently less active than the dark-fed group over time. Light-fed mice also consumed slightly more calories than the dark-fed mice. The combined effect of a small increase in caloric intake and a small decrease in activity contributes to specific differences in body weight.
- B) **FALSE.** Checked Mice are nocturnal animals with higher activity during the dark period (Figure 1) and less temperature fluctuation (Figure 2B). However, light-time feeding resulted in an increase in body weight and fluctuations in

body temperature. Therefore, light-time feeding is an example of circadian-behavior misalignment.

- C) **FALSE.** Checked The light-fed mice's body temperature range before the experiment was 36-38 and during the light-time feeding, it became 34-38 degrees Celsius (Figure 2A), while the dark-fed mice's body temperature range did not change (Figure 2B).
- D) **FALSE.** Checked From Figure 1, it can be seen that with decreasing locomotor activity in light-fed mice, their caloric intake from the middle of week 2 to the middle of week 6 stays relatively the same.

Task #41.

- A) **TRUE.** Checked The rate of active bouts persisted through prolonged periods of constant light or darkness (Figure 1C), suggesting internal control or circadian rhythm.
- B) **FALSE.** Checked From the figures it is seen that they are nocturnal animals because active and quiet sleep phases occur during the daytime.
- C) **FALSE.** Checked There is no data provided that shows the changes in the average duration of active bouts after sleep deprivation.
- D) **TRUE.** Checked There is an increased amount of bouts after sleep deprivation and there is a higher probability of bout being sooner if the bout is interrupted.

Task #42.

- A) **FALSE.** **checked** The electric eel employs low-voltage impulses to detect the presence of prey nearby. However, for direct localization of prey, it utilizes high-voltage impulses. This is evident from the eel's behavior of attacking the conducting rod, mistaking it for prey, as the rod affected the electric field generated by the high-voltage impulses.
- B) **FALSE.** **checked** The eel initiated the attack on the rod because it detected the presence of prey nearby. It recognized this when the prey, in response to the low-voltage impulse, twitched, thereby creating a mechanical movement in the water, which the eel sensed.
- C) **FALSE.** **checked** The frame indicates the time point when the high-voltage impulses commence (following the low-voltage one). From this, it can be inferred that the corpse has already twitched, and the eel has realized the presence of prey nearby.
- D) **FALSE.** **checked** As stated in the explanation of the first point, for direct prey localization prior to attack, the electric eel employs high-voltage impulses. Mechanoreceptors only sense water vibrations after the prey twitches due to the low-voltage impulse.

Task #43.

- A) **FALSE.** **Checked** Not all bones are represented by dark areas. For example, the eye sockets are shown in light gray.

- B) **FALSE. Checked** Conversely, as the passage widens, the airspeed decreases - Poiseuille's Law.
- C) **FALSE. Checked** Structure "I" is tongue (lingua), which has other primary functions
- D) **FALSE. Checked** Due to the large weight load per unit area of support, saigas more often fall into the snow, which makes them easier prey for wolves in winter

Task #44.

- A) **FALSE.** checked. Activation of TLR7/8 receptors on X-chromosome bearing sperm cells decrease their progressive motility (See Graph A of Figure 2).
- B) **TRUE.** checked. See Graph C of figure 2. High-mobility sperm cells that don't have TLR7/TLR8 receptors (and more likely to have Y chromosome) are in the upper layer.
- C) **TRUE.** checked. Results shown in graphs B and C of Figure 2 allow us to conclude that this is a viable method of sorting sperm cells by sex.
- D) **TRUE.** checked. Difference between R and C groups under different glucose concentration from Graph A of Figure 2

Task #45.

- A) **TRUE.** checked. Cross - sections of samples with EC(+) shows luminal structures and better media supply, which shows that EC increases the formation of new vessels, while function of FGF2 is to promote communication with the vascular bed.
- B) **FALSE.** checked. As you can see in the figures H and G with green fluorescence, the green dots are not actively distributed in the cell sheet area in image G, while at image H, EC are actively distributed around the cs.
- C) **TRUE.** checked. By assuming that cells have the same proliferation rate (from the same culture of cells), the plateau state in the bioluminescence rate indicates the cell survival rate, because at the plateau point, the number of living cells with luciferase enzyme is the only limiting factor. It is true that EC(-) FGF(+) has higher cell survival than EC(+) FGF(-).
- D) **FALSE.** checked. For EC(+) FGF(+) the rate can not mainly be explained by simple direct diffusion, because direct diffusion alone can not be that fast as you can see in the example of all EC(-) samples. The combined effect of direct diffusion and transport by newly developed vessels provides a fast increasing state.

Task #46.

- A) **FALSE.** (Checked) We can see that neurons can fire action potential with higher frequency (Figure 1C) meaning that this pattern is formed due to different reasons.

- B) **FALSE (Checked)** Since the membrane potential rises for the fourth time, the neuron is not in the absolute refractory period, therefore, with an increase in the signal duration, more ions will enter the cell and the potential may reach the threshold value.
- C) **TRUE (Checked)** The potential of sodium is much further from its equilibrium potential than the potential of potassium, therefore, even with the free movement of both cations, more sodium will enter the cell.
- D) **TRUE (Checked)** For the generation of action potential any duration of light pulse longer than certain threshold can work, however to get subthreshold depolarization duration of light pulse should be more precise.

Task #47.

- A) **FALSE (Checked)** Flying animals require more energy for the beginning of the movement.
- B) **FALSE (Checked)** It is possible to find the lowest value by solving an equation or taking the derivative. $V = 0.79 \text{ km/h}$.
- C) **FALSE (Checked)** Since each type of gait has its minimum, animals would switch between these COT minimums without moving on the intermediate speeds - multimodal distribution (with 3 peaks).
- D) **TRUE (Checked)** Cod experiences increasing drag because of the movement in the water, running animals experience negligible air drag.

Task #48.

- A) **TRUE (Checked)** The person does full expiration and the only remaining volume is RV, meaning that the spirometer is now $V_s + ERV$
- B) **FALSE (Checked)** From the description of the process, it is possible to notice that we form a closed system when a person starts to breathe in a spirometer. It means that the amount of helium at the beginning of the experiment should be equal to the final amount: $C_1 V_s = C_2 (V_{sf} + RV)$. As a result, we can calculate $RV = (C_1 V_s) / C_2 - V_{sf}$. $RV = (10 \cdot 5) / 8 - 5.6 = 0.65 \text{ L} = 650 \text{ ml}$
- C) **TRUE (Checked)** Constant inspiratory volume + increased RV = increased lung volume.
- D) **TRUE (Checked)** Patient 2 has decreased RV and decreased ERV, meaning that its total lung volume is also smaller with less RV left inside, pneumothorax can explain this as the patient would only have one functioning lung.

Task #49.

- A) **TRUE (Checked)** Tachycardia is increased heart rate, meaning that the time between each heartbeat, and consequently systole, is smaller.
- B) **FALSE (Checked)** High arterial pressure means that there is higher pressure in the aorta during the systolic phase and more resistance to ventricular contraction. Higher resistance results in lower peak velocity.

- C) **TRUE**. checked. By simply rearranging the equation, we get that $v = c(\Delta f) / (2f_0 \cos(\theta)) = 120 \text{ cm/s}$
- D) **TRUE**(Checked). The area under the curve represents the distance traveled by the blood during the entire systole. Blood travel distance = $0.33 * 100 / 2 = 16.67 \text{ cm}$. We know the distance and we know that the aorta has a radius of 1.25 cm , meaning that the volume traveled through the aorta is $16.67 * (\pi * 1.25^2) = 81.8 \text{ cm}^3$. From the graph, it is also seen that the heart rate is 60 bpm , $\text{CO} = 81.8 * 60 = 4908 \text{ cm}^3 = 4.9 \text{ L}$.

Task #50.

- A) **FALSE**(Checked). While hydrostatic pressure drives blood through the glomerulus, water moves via secondary active transport after the sodium.
- B) **TRUE**(Checked). While at the same depth, 1 and 3 have different osmolarities, this is not due to the water movement, but rather because of the reabsorption of the sodium in the collecting duct.
- C) **FALSE**(Checked). Osmolarity should be higher than 290 so that water would establish 290 in the descending limb, as well as in the collecting duct.
- D) **TRUE**(Checked). In this case, sodium will be left in the kidney and gradually all of the sodium will accumulate in the medulla.